

Dietary Cholesterol and Heart Disease

A Healthy Skepticism Case Study

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The advice on eggs keeps flipping. For fifty years, patients were told to limit them because eggs are full of cholesterol. A 2019 headline said eggs raise your risk of heart disease by 17%. By 2025 the headlines said the cholesterol scare was a myth.

A real claim rarely comes one fault at a time. The egg story makes four claims, and each carries a different fault.

Sources: Northwestern University news release, March 2019. ScienceDaily, July 2025, reporting Carter S et al.

The cholesterol you eat and the cholesterol in your blood

The old advice came from the idea that you are what you eat. But the cholesterol on your plate is not the cholesterol in your blood test. Your liver makes cholesterol, and when you eat more, it makes less.

Your LDL, often called bad cholesterol, depends on how fast your liver clears it from your blood. Saturated fat slows that clearing more than eaten cholesterol does.

Eating cholesterol raises LDL only a little. In feeding trials, where researchers control the diet, each egg a day adds about 4 to 9 mg/dL. Past two or three eggs a day, more eggs add little.

Sources: Lecerf J-M, de Lorgeril M. Br J Nutr 2011;106:6–14. Vincent MJ et al. Am J Clin Nutr 2019;109:7–16. Hopkins PN. Am J Clin Nutr 1992;55:1060–1070. USDA FoodData Central, egg, whole, raw.

What holds up: LDL and heart disease

The other half of the old idea holds up. LDL causes heart disease. Trials of drugs that lower LDL show it. Take 1,000 people at low risk and lower their LDL by about 39 mg/dL with a statin for five years. About 11 of them are spared a major heart or stroke event.

An LDL drop as big as a diet could ever produce spares few people, even at very high risk. In the IMPROVE-IT trial, 18,144 patients after a recent heart attack or unstable angina took a statin plus either ezetimibe or a placebo. Their LDL differed by about 16 mg/dL. Over seven years, 327 in 1,000 on ezetimibe had a major event, against 347 in 1,000 on placebo. That's about 20 fewer in 1,000. The margin of error on that result came close to including no benefit at all.

Sources: Cholesterol Treatment Trialists' Collaboration, cttcollaboration.org, people at low risk. Cannon CP et al. N Engl J Med 2015;372:2387–2397.

Claim 1. "Eggs raise your risk of heart disease by 17%"

The fault: relative reporting.

The course question it echoes: What does a 36% improvement from taking a drug really mean?

The press release gave the 17% and no counts. The number came from a 2019 JAMA study that followed 29,615 US adults, typically for 17.5 years, and recorded what they ate. Each extra 300 mg of dietary cholesterol a day, about one and a half eggs, went with 17% more cardiovascular disease.

A 17% rise doesn't tell you how many people it affects. That depends on how many would get the disease anyway. In this study, 5,400 of the 29,615 got cardiovascular disease over 17.5 years, about 180 in 1,000. Among people eating an extra 300 mg of cholesterol a day, about 32 more in 1,000 got it. Allowing for the margin of error, the true number could be anywhere from 14 to 51 in 1,000.

Sources: Zhong VW et al. JAMA 2019;321:1081–1095. Northwestern University news release, March 2019.

Claim 2. "Eggs cause heart disease"

The fault: the causal leap.

The course question it echoes: Does bird watching prevent dementia?

The claim draws an arrow from a correlation. Every study linking eggs to heart disease compares people who eat eggs with people who don't. No trial has assigned people to eat eggs or dietary cholesterol and counted heart attacks or deaths.

Something other than the eggs could make the link. Egg eaters differ from other people in what else they eat and how much they earn. Researchers call these differences confounders and try to correct for them in the math. The answer depends on what they correct for. In the JAMA study, the egg link disappeared once they corrected for total dietary cholesterol.

The studies don't agree. Some large US studies lean toward harm. Others, in the US and China, found no link or found that egg eaters did better.

"Eggs are safe" makes the same leap in reverse. It erases the arrow because the correlation is weak. But confounders can hide a real effect too.

Sources: Qin C et al. Heart 2018;104:1756–1763. Drouin-Chartier J-P et al. BMJ 2020;368:m513. Zhong VW et al. JAMA 2019;321:1081–1095.

Claim 3. "Keep cholesterol under 300 mg a day"

The fault: arbitrary categories.

The course question it echoes: How concerned should you be if your systolic blood pressure goes from 119 to 120?

I could not find where the 300 mg limit came from. It was in American Heart Association advice by 1968. It passed into the 1977 Senate Dietary Goals and later US Dietary Guidelines.

A line like this suggests something changes when you cross it. Nothing does. In feeding trials, LDL rises gradually with each egg and then levels off. The newer advice has the same problem. The American Heart Association's 2019 advisory allows about one egg a day for healthy people. No study finds that risk changes at one egg.

Sources: American Egg Board, "The evolution of dietary cholesterol recommendations." Nestle M. Adv Nutr 2018;9:148–150. Carson JAS et al. Circulation 2020;141:e39–e53. Hopkins PN. Am J Clin Nutr 1992;55:1060–1070.

Claim 4. Trials show eggs do and do not raise cholesterol

The fault: the Bernoulli fallacy.

The course question it echoes: Why does the published medical advice keep changing?

The trials only look contradictory. One review, from a group that advocates plant-based diets, found that 93 in 100 independently funded feeding trials showed eggs raising LDL. Then the conference abstract of a 2025 Australian trial was titled "Eggs do not raise blood cholesterol."

That title reads a p-value as proof of no effect. Forty-eight people completed three diets. Saturated fat clearly raised LDL. For cholesterol, the p-value was 0.42. The p-value is $P(\text{data} \mid \text{the hypothesis is false})$. Here the hypothesis is that cholesterol raises LDL. A p of 0.42 says data like these are quite likely if cholesterol had no effect.

The trial's own numbers fit the small effect the other trials found. Two of the diets differed only in cholesterol, 300 mg against 600 mg a day. Their LDL differed by about 3 mg/dL. That's about what the other feeding trials predict. Forty-eight people are too few to see an effect that small.

The official advice changed more than the evidence did. US guidance went from 300 mg a day, to "as little dietary cholesterol as possible" in 2015, to no mention at all in the 2025-2030 guidelines. The feeding evidence on LDL barely changed across those years.

Sources: Carter S et al. Am J Clin Nutr 2025 (corrigendum published). Carter S et al. Proc Nutr Soc 2025, conference abstract. Dietary Guidelines for Americans 2015–2020 and 2025–2030. Barnard ND et al. Am J Lifestyle Med 2021 (online 2019).

What cutting dietary cholesterol could do

No study has measured what cutting dietary cholesterol does to heart disease. We can only estimate it from the two findings above. Eaten cholesterol raises LDL a little, and lower LDL means fewer heart events.

The estimate is small. Cutting about 300 mg a day, one and a half eggs, lowers LDL by roughly 6 to 14 mg/dL. That's less than the 16 mg/dL gap in IMPROVE-IT, which spared 20 in 1,000 very high-risk patients over seven years. For most people the benefit would be smaller. It may be none.

Even that estimate takes a leap. It assumes LDL lowered by diet acts like LDL lowered by a drug. No one has tested that. The harm also tracks the number of LDL particles, which the usual blood test doesn't count. In one trial of 42 older adults, the particle count changed little, even in the people whose LDL rose most. That trial was too small to tell a small change from none.

Cutting cholesterol from food usually cuts saturated fat too, and saturated fat moves LDL more.

Sources: Vincent MJ et al. Am J Clin Nutr 2019;109:7–16. Cannon CP et al. N Engl J Med 2015;372:2387–2397. Borén J et al. Eur Heart J 2020;41:2313–2330. Greene CM et al. Nutr Metab 2006;3:6.

Summary

Claim	Fault	What the evidence shows
"Eggs raise your risk of heart disease by 17%"	Relative reporting	About 32 more in 1,000 over 17.5 years for each extra 300 mg a day, in observational studies
"Eggs cause heart disease"	The causal leap	No trial has tested it. The studies disagree and can't separate the egg from the eater
"Keep cholesterol under 300 mg a day"	Arbitrary categories	No threshold has been found. LDL rises gradually with intake and levels off
Trials show eggs do and do not raise cholesterol	The Bernoulli fallacy	Most trials find a small rise. The trial that found none was too small to see it

Two faults don't appear here. Probability reversal needs a diagnostic test read backwards, and this story has none. Claim 4 comes close, since reading a p-value as the chance of no effect is also a backwards reading. The syndrome trap needs one diagnosis covering different conditions. LDL is only a measurement. Spotting an absent fault is part of the skill.

What to ask

Relative reporting. Out of how many people, and over how long?

The causal leap. Was anyone assigned to eat the eggs, or were eaters compared with non-eaters?

Arbitrary categories. Where did the line come from, and what changes at it?

The Bernoulli fallacy. Was the study big enough to see the effect it says isn't there?

Educational, not medical advice.